



NEXT WAVE (INDIA)

SITAGLIPTIN TABLETS USP 50 mg

PART-II

P8.2 Post Approval Stability Protocol and Stability Commitment

Next Wave (India) hereby commits that a minimum one batch of Sitagliptin Tablets USP 50 mg every year shall be kept for stability studies for all years when manufacturing of the drug products is undertaken. Sample from the batch will be stationed at a long term storage conditions as per the stability requirements.

If any results fall outside of the retest / shelf-life specifications, these must be report together with the proposed action.

-----Stability protocol attached herewith----



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(HP) – 173025

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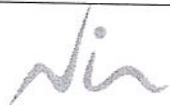
STABILITY STUDY PROTOCOL

Product Name	Sitagliptin Tablets USP 50mg	Strength	50 mg
Protocol No.	SP/NFRD/036	Effective Date	02/02/2022
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STABILITY STUDY PROTOCOL

Product Name	Sitagliptin Tablets USP 50 mg		
Protocol Number	SP/NFRD/036		
Revision Number	00		
Composition	Each film coated tablets contains: Sitagliptin Phosphate USP Eq. to Sitagliptin 50 mg Excipients q.s. Approved Colour Used		
Shelf Life	Proposed 36 months		
Dosage Form	Oral Solid		
Manufactured at	Next Wave (India)		
Stability Details	Type of Stability Study	Storage Condition	Period
	Accelerated	Temperature: 40°C ± 2°C Relative Humidity: 75% ± 5%	06 months
	Long Term	Temperature: 30°C ± 2°C Relative Humidity: 75% ± 5%	36 months
	Long Term	Temperature: 30°C ± 2°C Relative Humidity: 65% ± 5%	36 months
	Long Term	Temperature: 25°C ± 2°C Relative Humidity: 60% ± 5%	36 months

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Protocol Approval:

Prepared by:

Name	Department	Designation	Signature & Date
Jay Prakash Chaudhary	R&D	Sign Officer	 01/01/22

Checked by:

Name	Department	Designation	Signature & Date
Samal Kothiyar	R & D	Executive	 01/02/22

Approved by:

Name	Department	Designation	Signature & Date
Pawan Kumar Yadav	R&D	Manager	 02/02/22

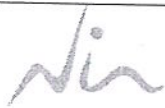
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	STABILITY STUDY PROTOCOL		
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1.0 Objective

To Design Stability Study Plan for **Sitagliptin Tablets USP 50 mg** manufactured by Next Wave(india), as per SOP No. R&D-004 "SOP for Stability Study "to assess Physico-Chemical and Microbiological Attributes when exposed to various storage conditions in its market pack and to verify its Shelf-Life and Storage Condition.

The objective of stability Study testing is to provide evidence on how the quality of drug product varies with time under the influence of a variety of environmental factors such as Temperature, Relative Humidity to establish the Shelf life for Drug Product and recommended Storage Conditions.

2.0 Scope

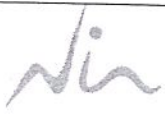
This Stability Protocol is applicable to conduct the stability study **Sitagliptin Tablets USP 50 mg at Next Wave (india)**.

This Protocol outlines the stability attributes & Specification, acceptance criteria, stability storage condition, test condition & Stability evaluation etc.

3.0 Responsibility

The following persons are involved in the stability study program & shall carryout the following responsibilities

Department	Responsibility
R&D Officer	- Charging of stability sample in chamber. - Monitoring of stability chamber & their storage condition. - Withdrawal of sample and report of same shall be maintained. - Analysis of stability sample
R&D Executive & HEAD R&D	- Review & approval of stability study protocol /report and analytical data. - Preparation of stability sample schedule & Protocol & Protocol report compilation of stability study data & prepare stability data summary report.

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4.0 Selection of Batches

Sample quantity of any subjected batch contains samples for following requirements:

- Sample required for one complete analysis for each time point
- Total quantity for all time points
- One additional pack for any unforeseen event.
- Placebo sample for each time point analysis

5.0 Sampling

The sample to be taken should be in market equivalent pack.

The sample quantity will be calculated to 1 pack extra in addition to the packs required for single analysis.

6.0 Storage Condition

The product manufactured will be subjected to 04 Condition:

- Accelerated (Temperature: 40°C ± 2°C, Relative Humidity: 75% ± 5%)
- Long Term Condition (Temperature: 30°C ± 2°C, Relative Humidity: 75% ± 5%)
- Long Term (Temperature: 30°C ± 2°C, Relative Humidity: 65% ± 5%)
- Long term (Temperature: 25°C ± 2°C, Relative Humidity: 60% ± 5%)

In case if stability chambers are out of order/under maintenance for more than 24 hours than this time will be included in the final testing of the batches. Such as if chamber is out of order for 2 days than these 2 days will be added at it's due date of testing for all subsequent time station.

7.0 Charging & Withdrawal Date

R&D officer /Executive person shall be charge & withdrawal required sample as per SOP R&D-004

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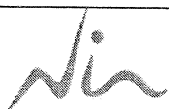
8.0 Product Details

Product Name	Sitagliptin Tablets USP 50 mg
Composition	Each film coated tablets contains: Sitagliptin Phosphate USP Eq. to Sitagliptin 50 mg Excipients q.s. Approved Colour Used
Batch No.	1) NW036A1 2) NW036A2 3) NW036A3
Mfg. Date	01/2022
Exp. Date	To be Determined (Proposed 36 months)
Batch Size	100000 tablets
Pack	Alu- Opaque PVC Blister (1X10 Tablets)

9.0 Stability Attributes & Specification

S.No.	Tests	Specification
1.0	Description	Light Beige , Round shape , film coated tablets
2.0	Identification By HPLC	The retention time of the major peak of the sample Solution corresponds to that of the standard Solution, as obtained in Assay.
3.0	Disintegration time	Not more than 5 minutes
4.0	Dissolution By HPLC	Not less than 75%(Q) labeled amount
5.0	Related Substance by HPLC (Procedure 1) Sitagliptin acid Fumarate adduct Sitagliptin triazecine analog Sitagliptin styrylacetyl analog	Not more than 0.2% Not more than 0.2% Not more than 0.2% Not more than 0.2%

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	Sitagliptin phenylcrotonyl analog	Not more than 0.2%
	Any other individual Impurity	Not more than 0.2%
	Total Impurities	Not more than 0.6%
6.0	Assay: Each Film Coated tablet contains: Sitagliptin Phosphate USP Eq. to Sitagliptin 50 mg	95.0% to 105.0% (47.5 mg to 52.5 mg)
7.0	Microbiological Examination Test	
	Total Bacterial count	Microbial count does not exceed 1000 cfu/g
	Total Yeast & Mould count	Microbial count does not exceed 100 cfu/g
	Pathogens: <i>Escherichia coli</i>	Absent in 1g

*Details of the specification and test methodology (as per Specification no. STP/NFRD/036-02) & STS No. STS/NFRD/036-02) which will be used for the analysis Sitagliptin Tablets USP 50 mg

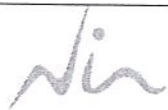
10.0 Acceptance Criteria

The product is considered to be stable if it passes all the criteria for the specifications mentioned above or there is no significant change.

Significant Changes:

- A 5% change in assay from its initial value, or failure to meet the acceptance criteria for potency. (Note: Other values may be applied, if justified, to certain products, such as multivitamins and herbal preparations).
- Failure to the acceptance criteria for related substance.
- Failure to meet microbial purity.
- Failure to meet the acceptance criteria for other parameters like description, Assay, Related substance, dissolution and any other characteristics.

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11.0 Stability Testing Schedule

S.No.	Tests	Initial	Type of Study							
			1M	3M	6M	9M	12M	18M	24M	36M
1.	Description	√	√	√	√	√	√	√	√	√
2.	Identification	√	√	√	√	√	√	√	√	√
3.	Dissolution	√	√	√	√	√	√	√	√	√
4.	Related Substance	√	√	√	√	√	√	√	√	√
5.	Assay	√	√	√	√	√	√	√	√	√
6.	Disintegration time	√	√	√	√	√	√	√	√	√
7.	Microbiological Contamination	√	X	X	√	X	√	√	√	√

Stability Condition	Test Stations	Types Stability
25°C & 60% RH	3 M, 6 M, 9 M, 12 M, 18 M, 24 M, 36 M,	Long Term
30°C & 65% RH	3 M, 6 M, 9 M, 12 M, 18 M, 24 M, 36 M,	Long Term
40°C & 75% RH	1 M, 3 M, 6 M	Accelerated
30°C & 75% RH	3 M, 6 M, 9 M, 12 M, 18 M, 24 M, 36 M,	Long Term

12.0 Documentation & Reporting

The R&D officer will enter the data obtained in the stability summary report and at the end of completion of stability study, a proper report will be made with conclusion,

All the analytical raw data carried out for stability will be retained in R&D along with COA made for each sample for each station.

13.0 Conclusion

After completion of stability study, conclusion about the assignment of shelf life to be drawn.

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14.0 References

SOP no.: R&D-004

15.0 Abbreviations

R&D : Research & Development
COA : Certificate of Analysis
SP : Stability Protocol

16.0 Annexures

All the entries of sample received for stability analysis shall be records as per "SOP R&D-004 mentioned in SOP for stability studies & stability study summary report as per Annexure VIII (SOP no. R&D-004. Format no. R&DF/004/08)

17.0 Revision History

Revision Number	Effective Date	Reason for revision
00	02/02/2022	Nil

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